1 Solution — GIAM 8.3 Problem 2

1.1 Problem

Can a diagonalization proof showing that the interval $(0,1)$ is uncountable be made workable in base-3 (ternary) notation?

1.2 Short Answer

Yes. Base 3 admits a workable diagonalization, but the substitution rule for picking the "complemented" diagonal digit needs more care than in base 10. The base-10 recipe (avoid 0 and 9) leaves 8 candidate digits, more than enough to differ from any single forbidden digit; the naïve base-3 analog (avoid 0 and 2) leaves only the single digit $\{1\}$, which is insufficient when the diagonal digit *is* 1. A small modification — alternating the digit set by parity of the position — fixes the issue.

1.3 The Subtlety — Why the Choice of Rule Matters

The diagonalization proof goes through three steps:

1. *Assume* $(0,1)$ is countable, so all reals in $(0,1)$ can be listed as $r_1, r_2, r_3, \ldots$ with a chosen base-b expansion $r_n = 0.d_{n,1}d_{n,2}d_{n,3}\cdots$ for each n.
2. *Construct* a new base-b sequence $0.e_1e_2e_3\cdots$ with $e_n \neq d_{n,n}$ for every n.
3. *Conclude* that the constructed sequence represents a real $r \in (0,1)$ that is not on the list, contradicting exhaustiveness.

The hidden hazard sits between steps 2 and 3. Some reals have *two* base-b representations:

$$\tfrac{1}{2} = 0.5000\cdots = 0.4999\cdots \quad \text{(base 10)},$$

$$\tfrac{1}{2} = 0.1000\cdots = 0.0111\cdots \quad \text{(base 2)},$$

$$\tfrac{1}{3} \;=\; 0.1000\cdots \;=\; 0.0222\cdots \quad \text{(base 3)}.$$

The general fact (mentioned but unproven in §8.3): in base b, a real $r \in (0,1)$ has *two* expansions iff $r = k/b^m$ for some positive integers k, m; the two expansions then differ by ending in trailing 0's vs. trailing $(b-1)$'s. All other reals have a unique base-b expansion.

Step 2 produces a *string* that differs from each listed *string*, but two different strings can represent the same *number*. If our constructed string happens to be the alternative representation of some r_n, the proof collapses: the strings differ, but the numbers don't.

The remedy is to constrain step 2 so the constructed string is the unique representation of its number — i.e., not a trailing-0's string and not a trailing-$(b-1)$'s string.

1.4 How Base 10 Handles It

In base 10, pick each e_n from $\{1,2,3,4,5,6,7,8\}$ — i.e., avoid 0 and 9 — and require $e_n \neq d_{n,n}$. Since the safe pool has 8 digits and at most one is forbidden by the diagonal constraint, there are always ≥ 7 legal choices.

The constructed real r has no 0 and no 9 anywhere in its expansion, so its expansion is neither a "trailing-0's" nor a "trailing-9's" representation. Hence r's base-10 representation is *unique*, and the constructed string is that unique representation. If r equalled some r_n, then r_n's listed representation would have to *be* our constructed string — contradicting $e_n \neq d_{n,n}$.

1.5 Why the Naïve Analog Fails in Base 3

Translate the base-10 recipe literally: pick e_n from $\{1, 2, \ldots, b-2\}$ avoiding 0 and $b-1$. In base 3, $b-2=1$, so the safe pool is the singleton $\{1\}$. With only one safe digit available, the rule "pick a safe digit different from $d_{n,n}$" is impossible whenever $d_{n,n} = 1$.

The naïve rule simply doesn't have enough room to maneuver. So we need a smarter rule.

1.6 A Working Rule for Base 3

Define e_n position-dependently, alternating which "safe pair" we draw from:

Parity of n	**If $d_{n,n} = 0$**	**If $d_{n,n} = 1$**	**If $d_{n,n} = 2$**
n odd	$e_n = 1$	$e_n = 2$	$e_n = 1$
n even	$e_n = 1$	$e_n = 0$	$e_n = 1$

Table 1.1

Equivalently:

$$e_n = \begin{cases} 1 & \text{if } d_{n,n} \neq 1, \\ 2 & \text{if } d_{n,n} = 1 \text{ and } n \text{ odd}, \\ 0 & \text{if } d_{n,n} = 1 \text{ and } n \text{ even}. \end{cases}$$

By inspection $e_n \neq d_{n,n}$ for every n. Structurally,

$$\text{odd } n: \quad e_n \in \{1, 2\}, \qquad \text{even } n: \quad e_n \in \{0, 1\}.$$

1.7 Verifying $r \in (0, 1)$ and Has Unique Base-3 Representation

Let $r = 0.e_1e_2e_3 \cdots$ in base 3.

1.7.1 $r > 0$

The first digit e_1 uses the odd rule, so $e_1 \in \{1, 2\}$. Hence $r \geq 0.1_3 = 1/3 > 0$.

1.7.2 $r < 1$

The second digit e_2 uses the even rule, so $e_2 \in \{0, 1\}$, in particular $e_2 \neq 2$. Hence the sequence is not all-2's, so $r < 0.222 \cdots_3 = 1$.

1.7.3 r's representation is unique

A real $r \in (0,1)$ has two base-3 representations iff $r = k/3^m$ for some positive integers k, m. The canonical (non-trailing-2's) representation of such a number terminates, i.e., has only finitely many non-zero digits. Conversely, a real with a non-terminating canonical representation has a unique base-3 representation.

The constructed sequence $0.e_1e_2e_3\cdots$ has:

- *Infinitely many digits* $\neq 2$: every even index n gives $e_n \in \{0,1\}$, and there are infinitely many even indices. So the sequence does *not* end in trailing 2's — it *is* the canonical representation of r.
- *Infinitely many digits* $\neq 0$: every odd index n gives $e_n \in \{1,2\}$, and there are infinitely many odd indices. So the canonical representation does not terminate.

Both conditions together rule out r being a finite fraction $k/3^m$. Hence r has a *unique* base-3 representation, and that representation is $0.e_1e_2e_3\cdots$.

1.8 Concluding the Contradiction

Suppose for contradiction that $r = r_n$ for some n. Then the listed representation of r_n is *some* representation of the number r. Since r has a unique base-3 representation, the listed string must equal our constructed string position-by-position. But by construction the listed n-th digit is $d_{n,n}$, while our constructed n-th digit is $e_n \neq d_{n,n}$. Contradiction.

So $r \in (0,1)$ but $r \notin \{r_1, r_2, r_3, \ldots\}$, contradicting the assumed exhaustiveness of the listing. Hence $(0,1)$ cannot be enumerated by $\mathbb{N}$, so $(0,1)$ is uncountable. ■

1.9 Spot Check — A Small Worked Example

Take the (made-up) listing

n	r_n **as a base-3 string**	$d_{n,n}$
1	$0.\mathbf{2}\,1\,0\,1\,2\cdots$	2
2	$0.0\,\mathbf{1}\,0\,2\,1\cdots$	1
3	$0.1\,2\,\mathbf{1}\,0\,1\cdots$	1

4	$0.2\,0\,1\,\mathbf{0}\,2\,2\cdots$	0
5	$0.1\,1\,2\,2\,\mathbf{2}\,0\cdots$	2
6	$0.0\,2\,0\,1\,1\,\mathbf{1}\cdots$	1

Table 1.2

Apply the rule: e_1 (odd, $d_{1,1} = 2$) $= 1$; e_2 (even, $d_{2,2} = 1$) $= 0$; e_3 (odd, $d_{3,3} = 1$) $= 2$; e_4 (even, $d_{4,4} = 0$) $= 1$; e_5 (odd, $d_{5,5} = 2$) $= 1$; e_6 (even, $d_{6,6} = 1$) $= 0$. So far

$$0.e_1e_2e_3e_4e_5e_6\cdots = 0.1\,0\,2\,1\,1\,0\cdots$$

This string differs from each r_n at position n, and (extending the rule indefinitely) has infinitely many odd-position digits in $\{1, 2\}$ and infinitely many even-position digits in $\{0, 1\}$ — so it is the unique base-3 representation of some $r \in (1/3, 1)$.

1.10 Remark — Alternative: Use Two-Digit Blocks ("Base 9")

Another clean fix: process the diagonal in *pairs* of base-3 digits. Each pair $(a, b) \in \{0, 1, 2\}^2$ takes one of 9 values, equivalent to a base-9 digit $3a + b$. The dual-representation pairs are $(0, 0)$ (the "trailing-0's" block) and $(2, 2)$ (the "trailing-2's" block) — by avoiding *both*, we ensure neither global trailing-0's nor global trailing-2's.

For each row index n, look at the diagonal pair $(d_{n,\,2n-1}, d_{n,\,2n})$ and replace it with any pair other than itself, $(0, 0)$, or $(2, 2)$ — there are at least $9 - 3 = 6$ safe choices, say the lexicographically first one. The resulting global expansion has no $(0, 0)$ block and no $(2, 2)$ block at any chosen position, hence is neither trailing-0's nor trailing-2's, hence is the unique representation of its number. The diagonal argument then proceeds as in the base-9 case directly.

This pairing approach is essentially "the base-10 argument done in base 9."

1.11 Remark — Why Base 2 Genuinely Resists the Single-Digit Approach

In base 2, the analog “avoid 0 and $b-1=1$” leaves the empty set — *no* safe digits. Worse, the only available rule (flip each diagonal bit) is precisely what fails, because the bit-flipped string can equal an alternate representation of a listed real. This is the obstacle the textbook flags in §8.3: the binary diagonal argument proves something *slightly different* (every listing of *binary strings* is missing some string), which translates to "$\mathcal{P}(\mathbb{N})$ is uncountable” rather than "$\mathbb{R}$ is uncountable” directly.

Even base 2 is salvageable, however, with the same pairing trick: process pairs as base-4 digits, where the safe pool $\{01, 10\}$ avoids both 00 and 11. That gives at most 1 disallowed pair (the diagonal pair), leaving ≥ 1 choice. So *any* base $b \geq 2$ admits a workable diagonal argument — with single digits for $b \geq 3$ (using a position-dependent rule for $b = 3$, freely for $b \geq 4$), and with digit-pairs for $b = 2$.

1.12 Remark — Where the Argument Sits Conceptually

The dual-representation issue is *not* a flaw in the diagonal argument; it is a feature of *positional notation*. The same real number has more than one name in any positional system, and a proof that quantifies over “all sequences of digits” needs to distinguish *sequences* from *numbers*. The fix — whether the base-10 “avoid 0 and 9”, the base-3 parity rule above, or the base-2 pairing — is always the same idea: *constrain the constructed string to land on a number with a unique representation.* Once that constraint is in place, “string differs at position n” upgrades cleanly to “number differs from r_n”, and the contradiction goes through.